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EXPERIMENT:-4

Aim: To understand Continuous Integration, install Jenkins, and build CI/CD pipelines using Declarative and Scripted syntax.

LO mapped: **LO3**

LO3: To Build and Manage automated CI/CD pipeline, for application development,packaging,testing and deployment

Theory:

Continuous Integration (CI)

Continuous Integration (CI) is a software development approach in which developers regularly merge their code into a shared repository. Each integration triggers automated processes such as building and testing the application, allowing issues to be identified and resolved early. This practice improves collaboration, reduces integration problems, and helps maintain software quality throughout the development cycle.

Jenkins

Jenkins is a widely used open-source automation server designed to support Continuous Integration and Continuous Delivery (CI/CD). It automates repetitive development tasks such as compiling code, running tests, generating reports, and deploying applications. With its extensive plugin ecosystem, Jenkins integrates with numerous development tools and platforms, making it suitable for a variety of software projects.

Jenkins Pipeline

A Jenkins Pipeline is an automated workflow that defines the sequence of tasks involved in building, testing, and deploying an application. It enables teams to represent the complete CI/CD process as code, ensuring that software delivery is consistent, repeatable, and easy to maintain.

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Advantages of Jenkins Pipeline:

- Automates the complete build, test, and deployment workflow.
- Provides a clear, stage-wise view of the execution process.
- Supports *Pipeline as Code*, allowing pipeline configurations to be version-controlled.
- Handles complex workflows through features such as parallel execution, conditional stages, and reusable scripts.

Types of Jenkins Pipelines

1. Declarative Pipeline

- Uses a structured and user-friendly syntax.
- Defined within a pipeline {} block containing stages and steps.
- Easier to read, maintain, and troubleshoot, making it suitable for most CI/CD implementations.

2. Scripted Pipeline

- Written using Groovy scripting within a node {} block.
- Offers greater flexibility and customization for advanced workflows.
- Best suited for projects that require complex logic or highly customized pipeline behavior.

Procedure

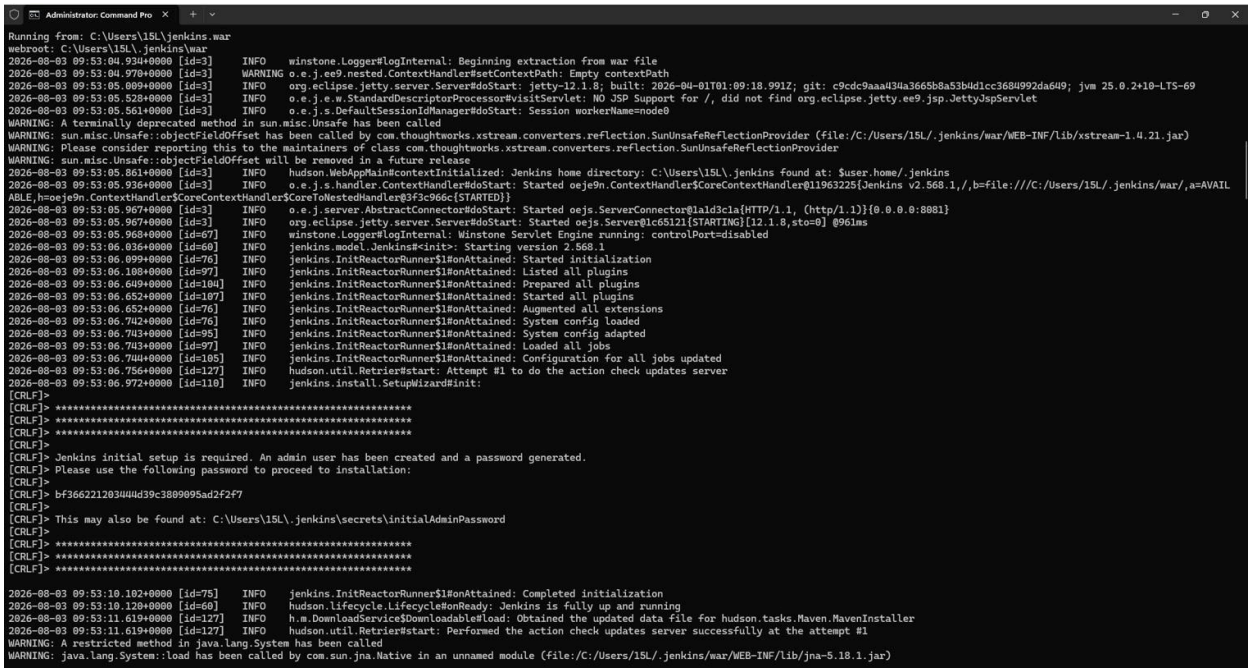
1. Verified that the Java Development Kit (JDK) was correctly installed and configured on the system.
2. Downloaded the jenkins.war file and started the Jenkins server using the command java -jar jenkins.war.
3. Opened Jenkins through http://localhost:8080, entered the generated administrator password, installed the recommended plugins, and completed the initial administrator setup.
4. Created two pipeline jobs:

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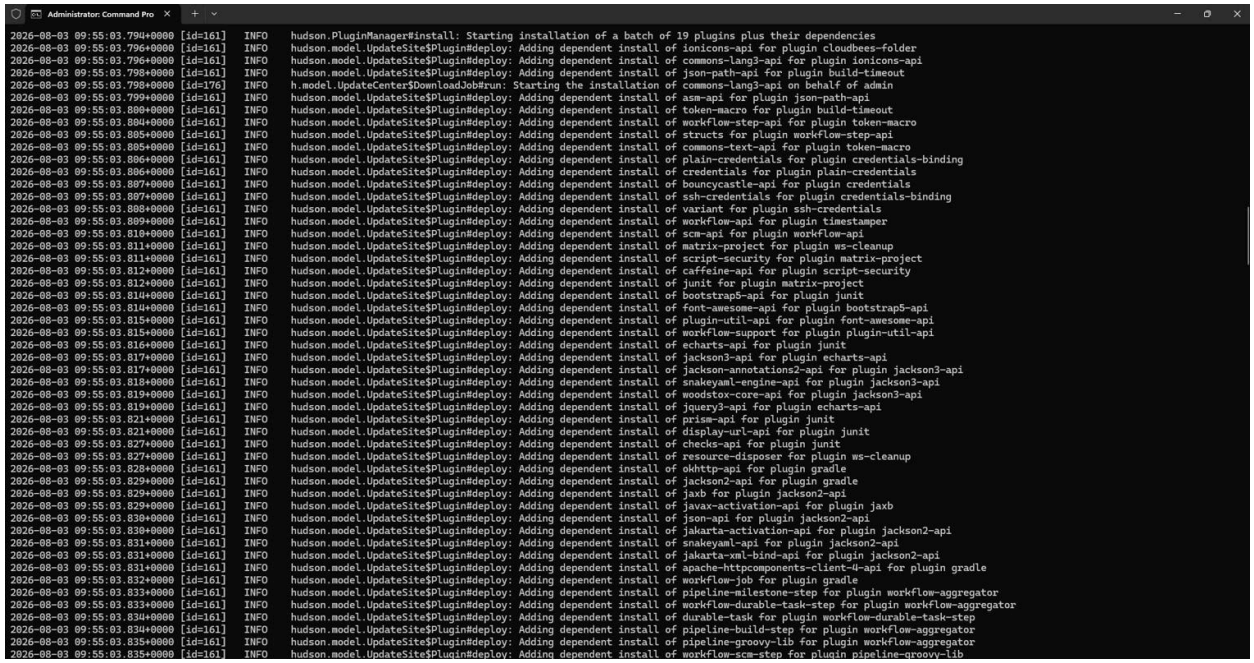
- A Declarative Pipeline using the pipeline {} syntax.
- A Scripted Pipeline using the Groovy node {} syntax.

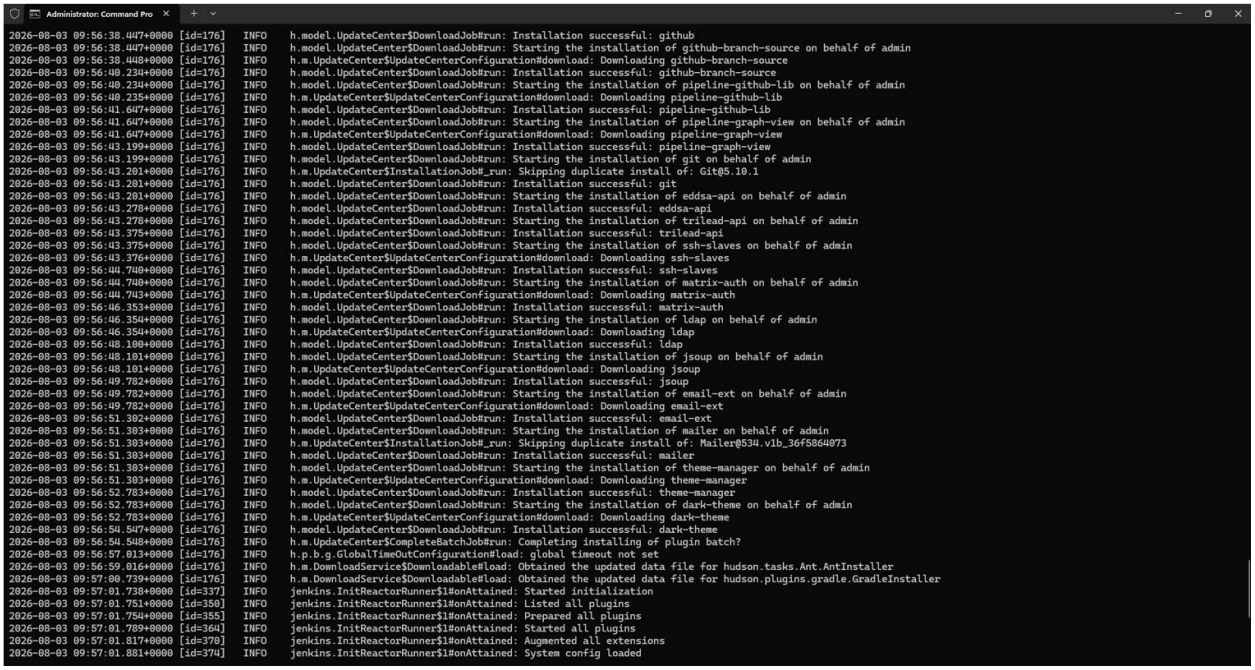
5. Executed both pipelines and confirmed their successful completion by reviewing the Console Output and build status in Jenkins.

Output:



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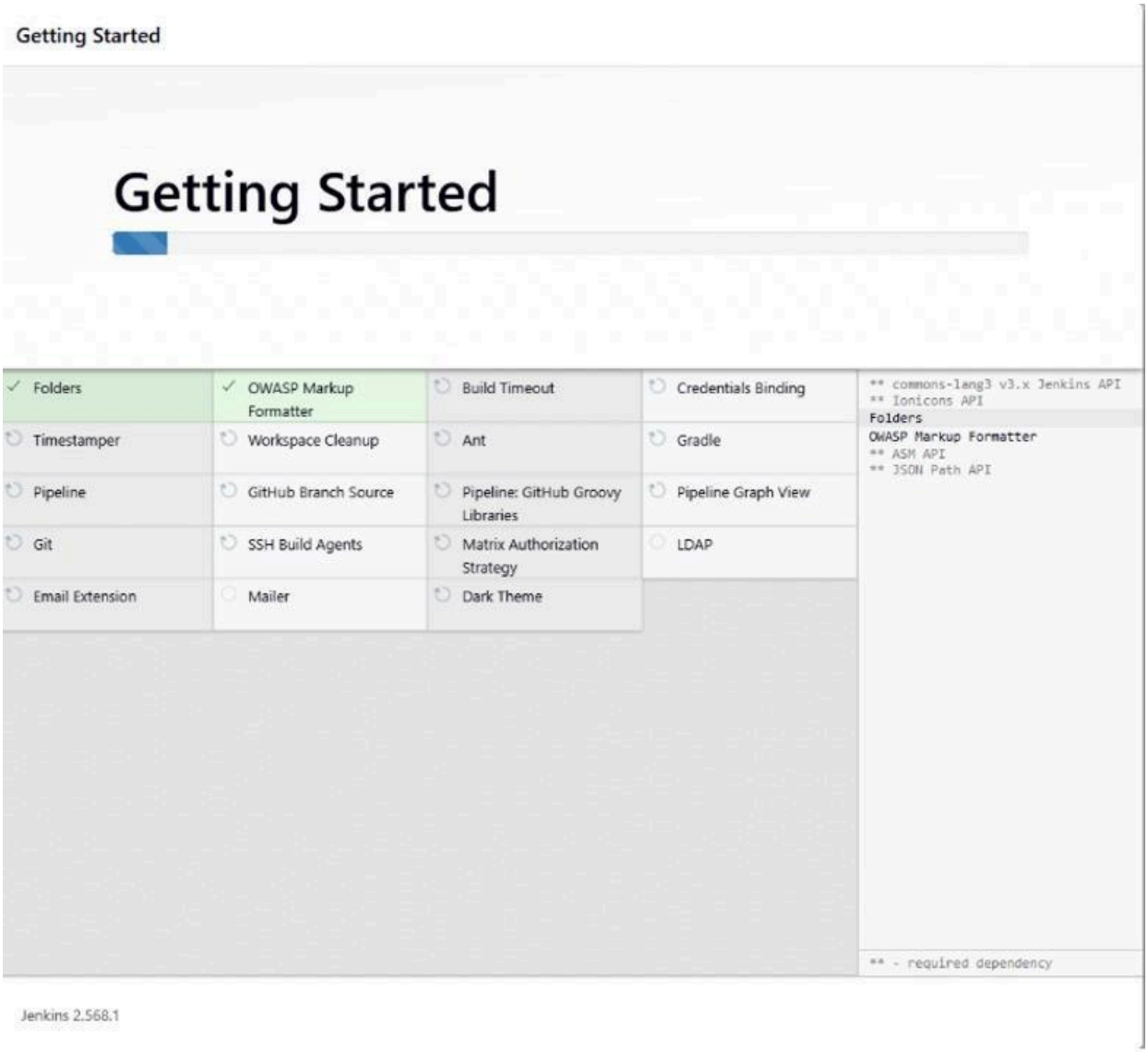




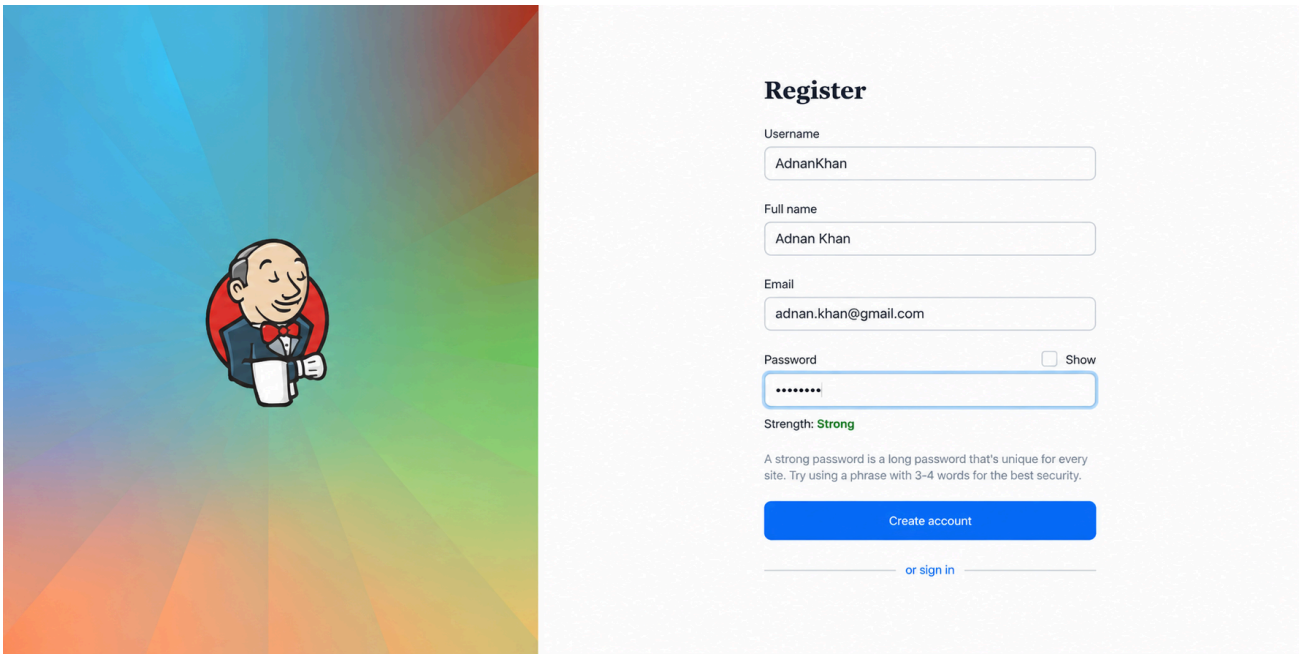
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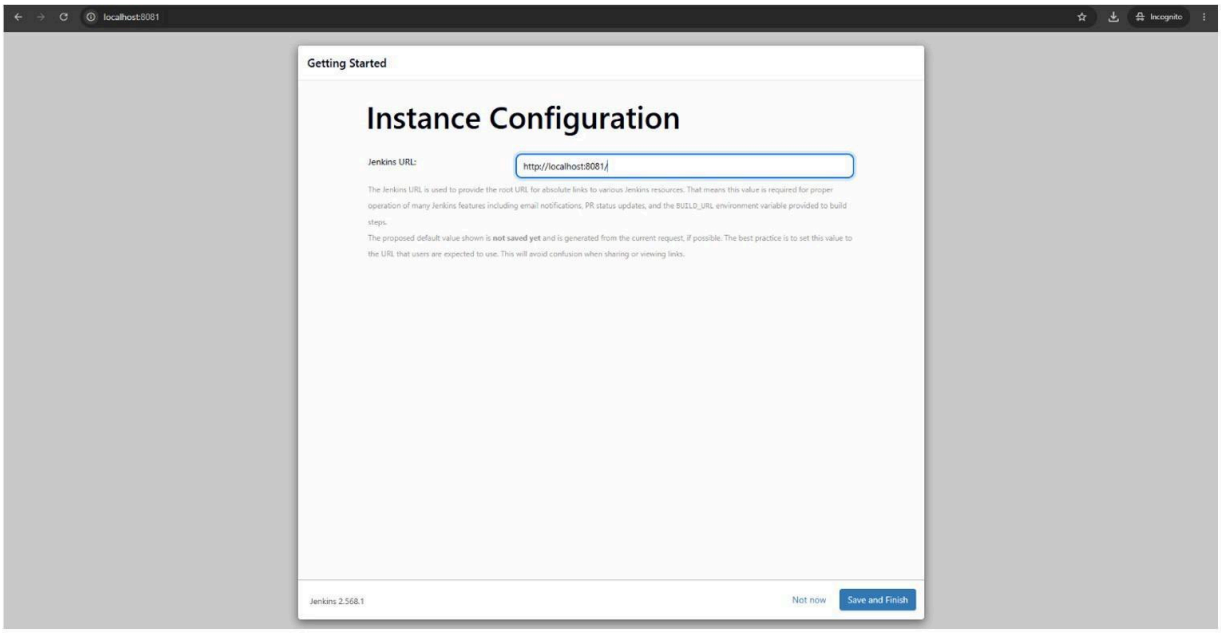


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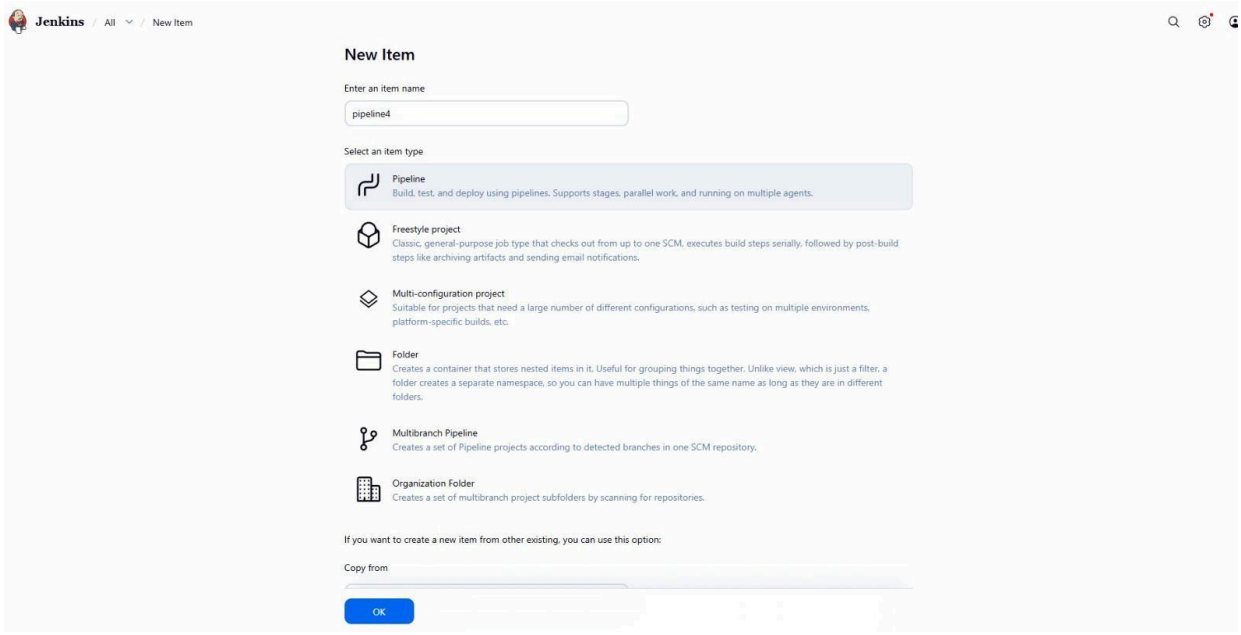


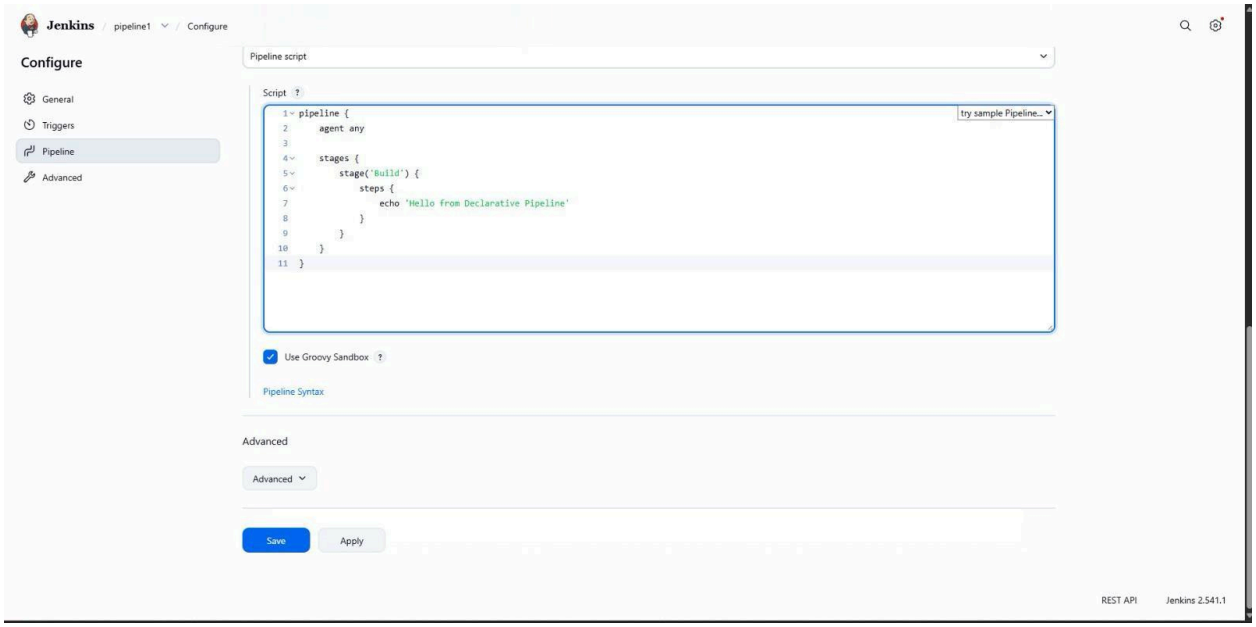


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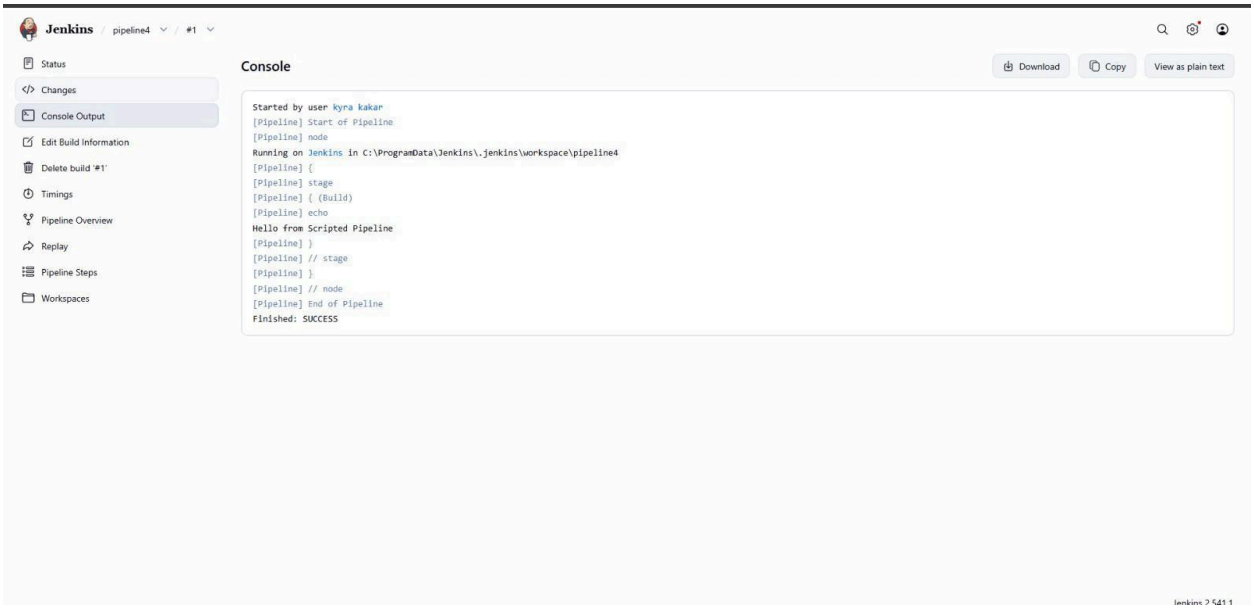


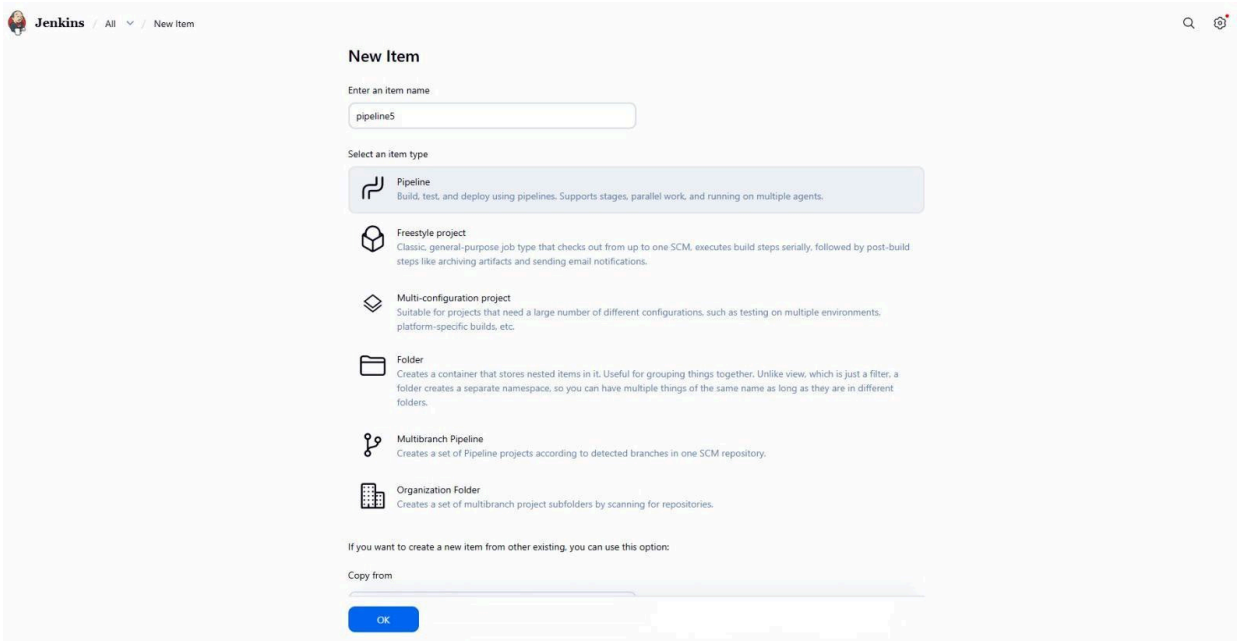
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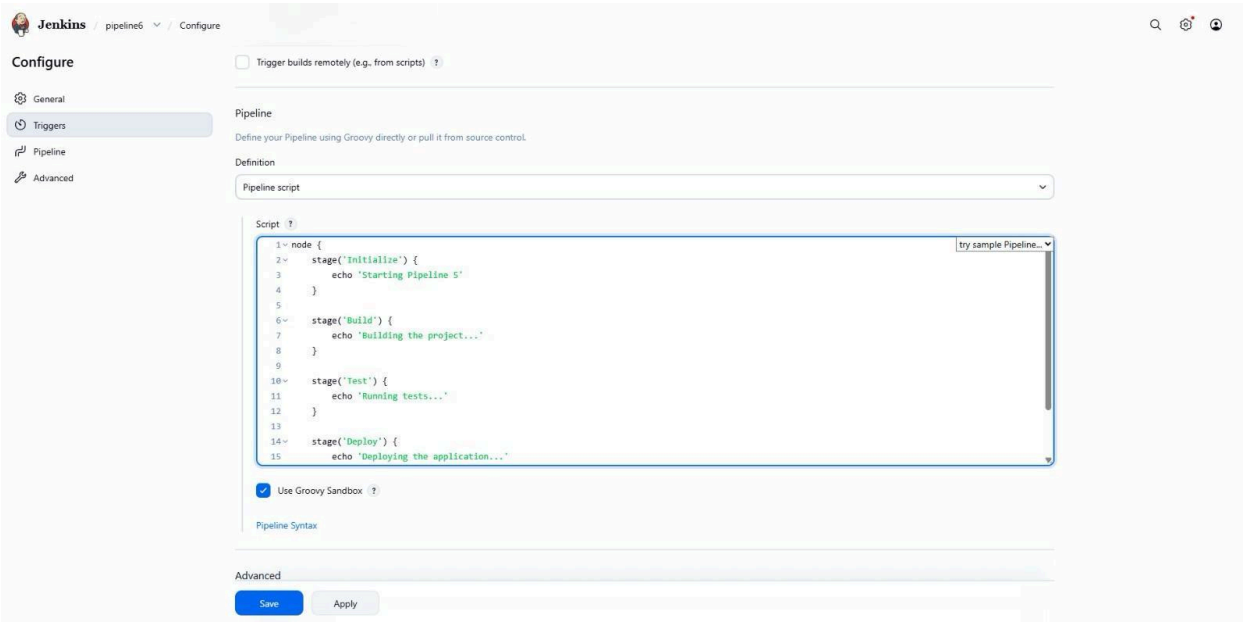


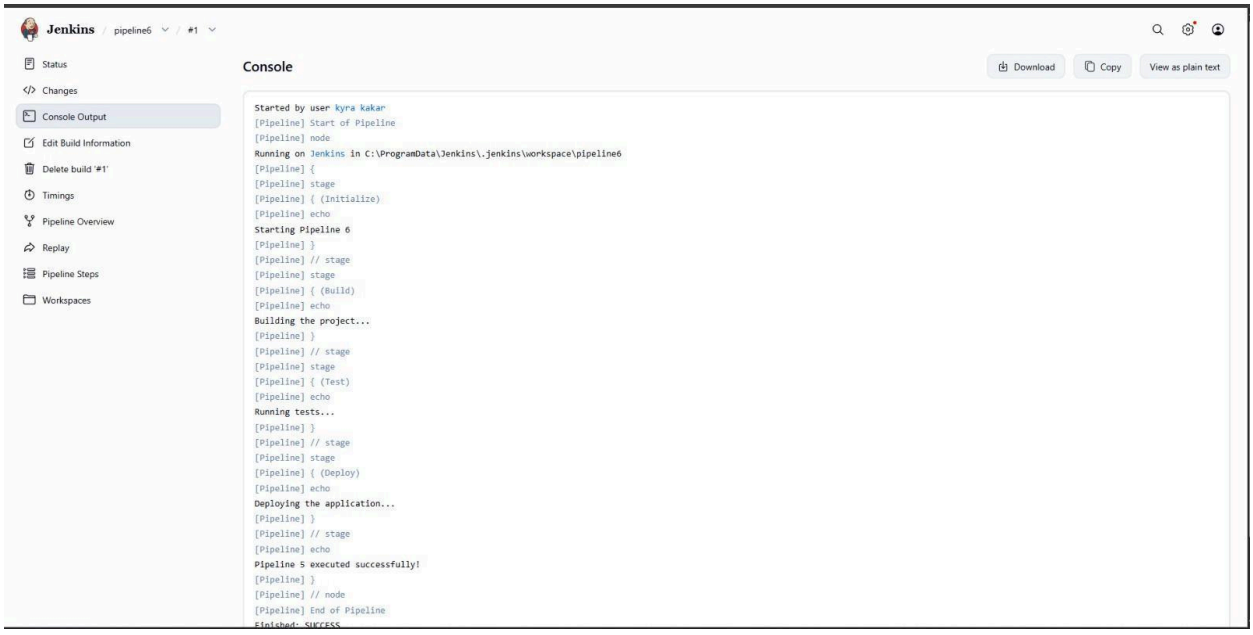
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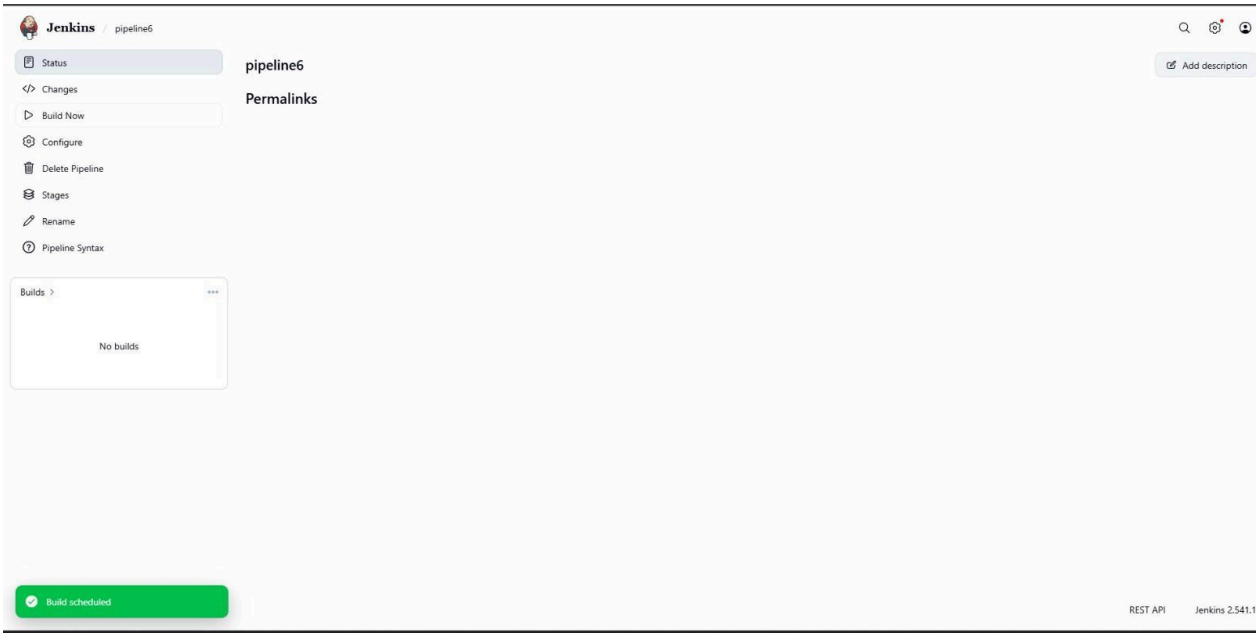


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Conclusion:

This experiment helped us understand the fundamentals of Continuous Integration (CI) and Continuous Delivery/Deployment (CD) using Jenkins. We successfully installed and configured Jenkins, verified its setup, and became familiar with its web-based interface. We also created and executed both Declarative and Scripted Pipelines, allowing us to understand their syntax, features, and practical use cases.

The experiment provided practical knowledge of how Jenkins automates key software development activities such as building, testing, and deploying applications. We learned how pipelines can be defined as code, making the CI/CD process more organized, reusable, and easier to maintain. Automation also demonstrated its benefits in reducing manual work, minimizing errors, and improving the speed and reliability of software delivery.

Overall, this experiment gave us hands-on experience with Jenkins and its role in implementing CI/CD practices. It strengthened our understanding of automated software development

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workflows and provided valuable skills that can be applied in real-world DevOps and software development environments.

LO3 mapped successfully.